

UPES (University of Petroleum and Energy Studies)

Department of Computer Science & Engineering

A Project Report On

FitGenie AI – Personalized AI Fitness & Nutrition Coach

An AI-Powered, Fully Responsive Web Application for Personalized Workout Planning, Nutrition Coaching & Progress Tracking

Submitted in partial fulfilment of the requirements for
B.Tech Summer Internship / Capstone Project

Submitted By	Submission Date
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Live Application: <https://fitgenie-ai.netlify.app>
Source Repository: <https://github.com/dhruv-jangid/fitgenie-ai>

Acknowledgement

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Dhruv Jangid

Abstract

The health and fitness technology space has grown rapidly, yet most consumer apps remain generic, static, and disconnected from a user's real-time condition, lifestyle, and goals. "FitGenie AI – Personalized AI Fitness & Nutrition Coach" addresses this gap with a fully responsive, browser-based platform that generates individualized workout and nutrition plans directly from a user's biometrics, goals, equipment, dietary restrictions, and injuries — built end-to-end with HTML5, CSS3, and vanilla JavaScript, without a heavyweight framework, so it stays lightweight and easy to deploy on static hosting.

A multi-step onboarding wizard captures age, gender, height, weight, goal, experience level, equipment, training frequency, diet style, allergies, sleep, hydration, activity level, injuries, and medical conditions. From this, the app computes BMR and TDEE in real time using the Mifflin-St Jeor method, derives a goal-adjusted calorie/macro target, and assembles a workout plan, nutrition and grocery plan, and progress dashboard. Differentiating features include interactive injury-adaptation, one-tap meal replacement with grocery sync, an AI coaching chat, gamified streaks/badges, and Chart.js-powered analytics.

This report documents the project's full lifecycle — problem identification, requirement analysis, design, implementation, testing, deployment, and reflection — using a genuine GitHub-to-Netlify workflow. The final deliverable is a fully working, publicly accessible application at <https://fitgenie-ai.netlify.app>.

1. Introduction

Fitness technology has grown rapidly, driven by rising health awareness and a shift toward data-driven living. Yet most fitness apps still rely on static templates and fixed workout libraries that place the entire burden of personalization on the user.

"FitGenie AI – Personalized AI Fitness & Nutrition Coach" reimagines the fitness-app experience as a personalized, browser-based "Health Operating System." Instead of a fixed catalogue of workouts and recipes, FitGenie AI collects biometric, lifestyle, and preference data through a guided onboarding wizard and uses it to computationally derive a workout structure, nutrition plan, hydration/sleep targets, and progress projection — rendered through a modern, glassmorphic, mobile-first interface.

The project was built with HTML5, CSS3, and vanilla JavaScript, enriched with Lucide icons, Chart.js, and canvas-confetti via CDN, avoiding a heavy build pipeline. It was version-controlled with Git/GitHub and deployed with continuous delivery on Netlify — an authentic, industry-style workflow. A responsive web app was chosen over a native app for zero-install accessibility, a single shared codebase across devices, and Netlify's free, instant deployment pipeline.

2. Problem Statement

Despite the large number of fitness apps available, most share a common set of shortcomings that reduce long-term effectiveness and user retention.

2.1 Limitations of Current Fitness Applications

- Generic, non-adaptive plans that do not account for a user's equipment or injuries.
- Manual, tedious food logging that leads to abandonment within days.
- No injury-awareness — users must manually guess safe exercise substitutes.
- Fitness and nutrition tracking split across disconnected apps or paid tiers.
- Weak motivation design, with little gamification or coaching presence.
- True personalization only available through expensive human trainers.

2.2 How FitGenie AI Solves These Issues

Problem in Existing Apps	FitGenie AI Solution
Generic, non-adaptive workout templates	18-step onboarding wizard computes a plan specific to the user's profile
Manual and tedious food logging	Auto-generated nutrition plan with 1-tap 'Replace Meal' and grocery sync
No injury-awareness	Interactive Injury Adaptation reroutes the plan away from affected joints

Problem in Existing Apps	FitGenie AI Solution
Disconnected fitness & nutrition	Single unified dashboard covering workouts, nutrition, and metrics
Weak engagement design	Gamified streaks, badges, and an AI Chat Coach
Expensive personalization	Delivered free through a purely computational, browser-based engine

3. Objectives

1. Design and develop a fully responsive web application across desktop, tablet, and mobile.
2. Build a multi-step onboarding wizard capturing biometric, lifestyle, and preference data.
3. Implement a real-time BMR/TDEE calculator that updates live as metrics are entered.
4. Generate personalized workout and nutrition plans from the user's profile.
5. Implement an interactive injury-adaptation feature for automatic plan adjustment.
6. Provide a Chart.js analytics dashboard for weight, consistency, and progress tracking.
7. Incorporate gamification (streaks, badges) and an AI-styled chat coach for motivation.
8. Follow an authentic Git/GitHub/Netlify development and deployment workflow.

4. Scope of the Project

4.1 In Scope

- A public landing page with features, pricing tiers, testimonials, and FAQs.
- Authentication screens (Sign In, Sign Up, Forgot Password) with validation.
- A 4-step onboarding wizard (Bio Profile, Goals, Lifestyle, Review) with a live BMR/TDEE Bio-Calculator.
- A dashboard, workout plan, nutrition plan with meal replacement, and grocery checklist.
- Hydration, sleep, and daily-metrics tracking with progress-tracking charts.
- Interactive injury management and an AI Chat Coach panel.
- Dark/light theme toggling and deployment to Netlify with a public URL.

4.2 Out of Scope (Current Version)

- A dedicated backend server or persistent cloud database (uses browser local storage).
- Real wearable-device integration (Apple Watch, Fitbit, Garmin).
- Real payment gateway processing for the displayed subscription tiers.
- Camera-based exercise-form detection or computer-vision meal recognition.

5. Existing System

A review of widely used fitness and diet-tracking apps was carried out to understand prevailing design patterns and their shortcomings.

5.1 Disadvantages of Existing Systems

#	Disadvantage	Impact on the User
1	Generic plans not tailored to equipment/injuries	Users perform unsafe/impractical exercises and quit
2	Manual food logging fatigue	Nutrition tracking abandoned within the first week
3	No real-time adaptation to pain/soreness	Increased injury risk and loss of trust
4	Fragmented fitness/nutrition/analytics modules	Confusing experience across disconnected screens
5	Shallow onboarding	Recommendations feel irrelevant
6	Weak motivational design	Low daily usage and high churn

6. Proposed System

FitGenie AI consolidates workout planning, nutrition planning, hydration tracking, injury management, analytics, and motivational coaching into one cohesive interface.

6.1 How FitGenie AI Improves the User Experience

- Deep, structured onboarding: an 18-step wizard builds a rich, individualized profile.
- Live feedback: BMR, TDEE, and target calories recalculate instantly as the user types.
- Computed, not browsed, recommendations assembled directly from the user's profile.
- Interactive injury adaptation reroutes the plan away from a flagged joint/muscle instantly.
- One-tap meal replacement with automatic grocery-list sync.
- Unified analytics dashboard and a gamified motivation layer (streaks, badges, confetti).
- AI Chat Coach and a consistent, premium glassmorphic visual design with theme toggling.

6.2 Advantages Over Existing Systems

Feature	Existing Systems	FitGenie AI
Onboarding depth	3–5 basic fields	18-step profile across 4 categories
Injury handling	Manual exercise substitution	One-tap interactive adaptation
Meal planning	Manual search & log	Auto-generated plan with 1-tap replace
Analytics	Basic totals	Trend charts + 12-week projection

Feature	Existing Systems	FitGenie AI
Access model	Native app install required	Instant browser access, zero install

7. Functional Requirements

ID	Requirement	Description
FR-1	Registration & Login	Users can sign up, sign in, and reset a forgotten password.
FR-2	Guided Onboarding	A 4-step wizard collects profile data needed to personalize the plan.
FR-3	Live BMR/TDEE Calculation	Recalculates BMR, TDEE, and target calories instantly as fields change.
FR-4	Workout Plan Generation	Generates a plan based on goal, level, equipment, and days/week.
FR-5	Nutrition Plan Generation	Generates a macro-based meal plan respecting diet style and allergies.
FR-6	Meal Replacement & Grocery Sync	Users can swap a meal; the grocery checklist updates automatically.
FR-7	Injury Adaptation	Selecting a sore joint/muscle triggers automatic workout rerouting.
FR-8	Metrics & Dashboard	Tracks hydration/sleep and summarises streaks and daily targets.
FR-9	Analytics & Gamification	Chart.js trend views plus streaks, badges, and celebration effects.
FR-10	AI Chat Coach & Theming	Conversational guidance panel and light/dark theme toggling.

8. Non-Functional Requirements

Category	Requirement
Performance	Pages load in under 2 seconds on standard broadband/4G via static hosting.
Responsiveness	Layout adapts cleanly across mobile, tablet, and desktop breakpoints.
Usability	Floating-label forms with inline validation and clear error states.
Availability	Publicly accessible via Netlify's CDN-backed hosting with minimal downtime.
Maintainability	Code organised into clear HTML templates, one stylesheet, and modular JS.
Portability	Runs consistently across Chrome, Firefox, Edge, and Safari.
Security	Client-side validation and maskable password fields.

9. Software Requirements

Component	Specification
Code Editor	Visual Studio Code (VS Code)
Version Control / Hosting	Git and GitHub
Deployment Platform	Netlify (continuous deployment from GitHub)
Browsers for Testing	Chrome, Firefox, Edge, Safari
Core Technologies	HTML5, CSS3, JavaScript (ES6+)
External Libraries (CDN)	Lucide Icons, Chart.js, canvas-confetti

10. Hardware Requirements

Component	Minimum Specification
Processor	Intel Core i3 / AMD Ryzen 3 or equivalent
RAM	4 GB (8 GB recommended)
Storage	256 GB with 2 GB free for project files
Display	1366×768 or higher
End-user device	Any smartphone, tablet, laptop, or desktop from the last 6–7 years

11. Technology Stack

FitGenie AI was built with foundational web technologies rather than a heavyweight framework, keeping it fast, dependency-light, and easy to deploy on static hosting.

11.1 HTML5

Provides the semantic structure of every screen. Separate `<template>` blocks for the Public (landing/auth), Onboarding, and App (dashboard, workout, nutrition) layouts are injected into a single `#app` root by JavaScript, avoiding the overhead of a router library.

11.2 CSS3

Defines the entire visual system in one `style.css` file — CSS variables for theming, glassmorphism (backdrop-filter blur), Grid/Flexbox for responsive layouts, and keyframe animations for micro-interactions and media queries for breakpoints.

11.3 JavaScript (ES6+)

Drives all interactivity in `app.js`: screen rendering, form validation, live BMR/TDEE calculations, the onboarding wizard, workout/nutrition plan generation, injury-adaptation logic, theming, and integration with Chart.js and canvas-confetti.

11.4 VS Code, GitHub & Netlify

VS Code was the primary IDE, with Git source control, IntelliSense, and linting extensions streamlining development. GitHub hosts the repository (dhruv-jangid/fitgenie-ai) and version history, and integrates directly with Netlify for continuous deployment — every push to main automatically rebuilds and republishes the live site.

11.5 Supporting Libraries

Library	Purpose
Lucide Icons	Lightweight SVG icon set used across the UI
Chart.js	Renders the analytics dashboard's trend charts
canvas-confetti	Celebratory animation on milestone completion

12. System Architecture

FitGenie AI follows a client-side, single-page-application-style architecture — there is no dedicated application server; all rendering, state management, and computation happen in the browser, with static assets delivered via Netlify's CDN.



Figure 12.1 – High-Level System Architecture of FitGenie AI

12.1 Architectural Layers

- **Presentation Layer (HTML Templates)**: reusable blocks for Public, Onboarding, and App screens.
- **Application Logic (app.js)**: navigation, form handling, validation, and orchestration.
- **Recommendation Engine**: computes BMR/TDEE and assembles workout/nutrition plans with injury adaptation.
- **Persistence**: browser local storage keeps the profile, plans, and logged metrics between sessions.

- Deployment Pipeline: GitHub-to-Netlify continuous deployment republishes the site on every push.

13. Flowchart

The flowchart below traces a first-time user's journey through the FitGenie AI application, from initial landing through to ongoing progress tracking.

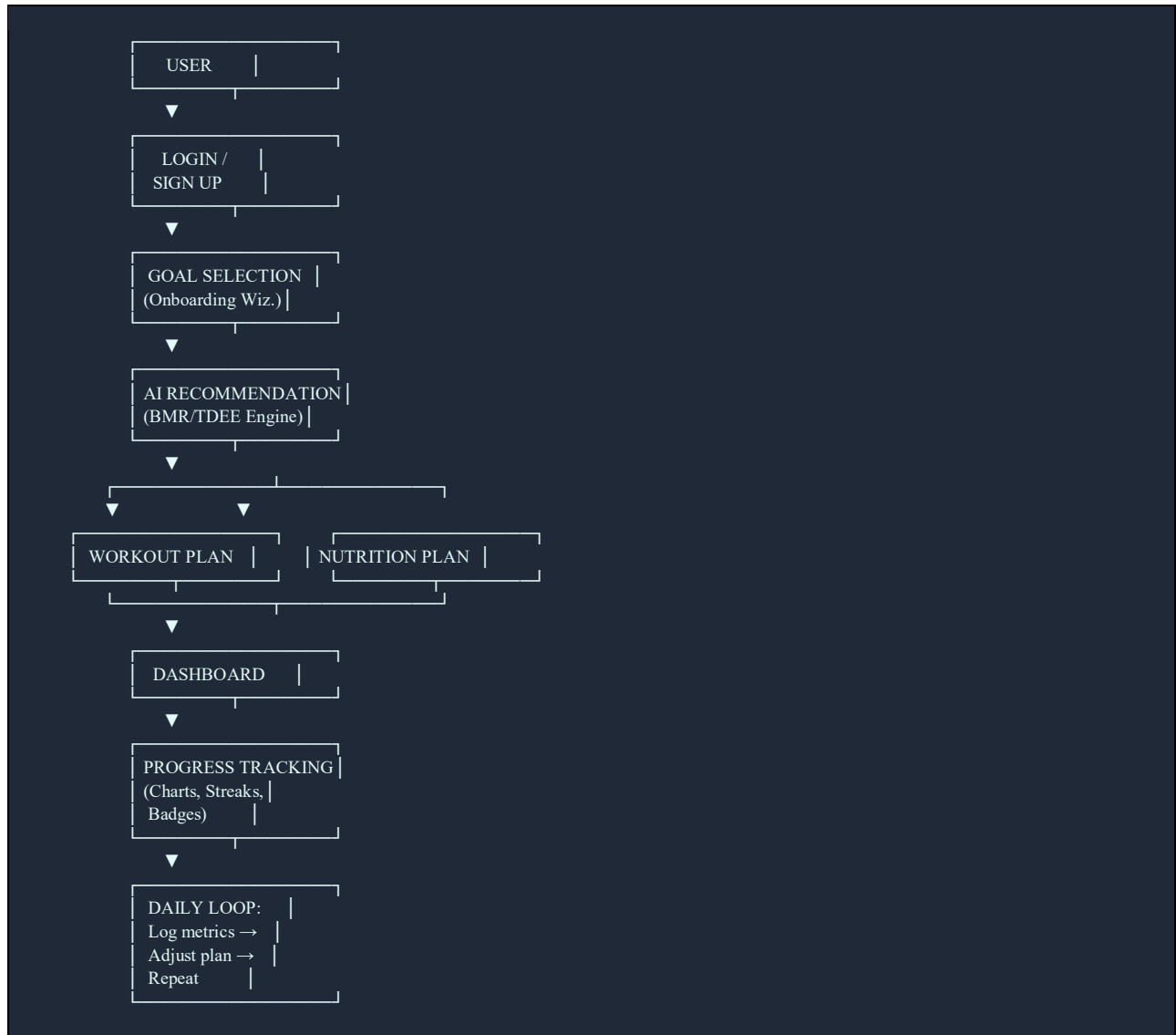
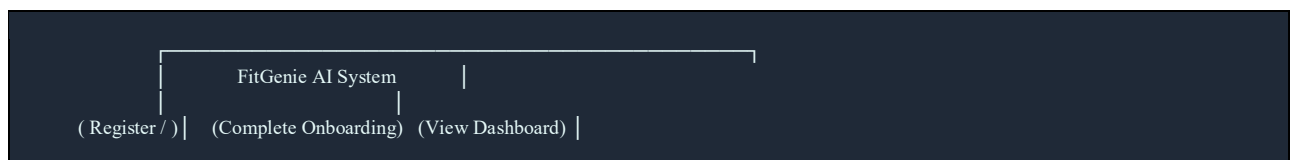


Figure 13.1 – User Journey Flowchart

14. Use Case Diagram

The use case diagram below identifies the primary actor (the App User) and the key use cases supported by FitGenie AI.



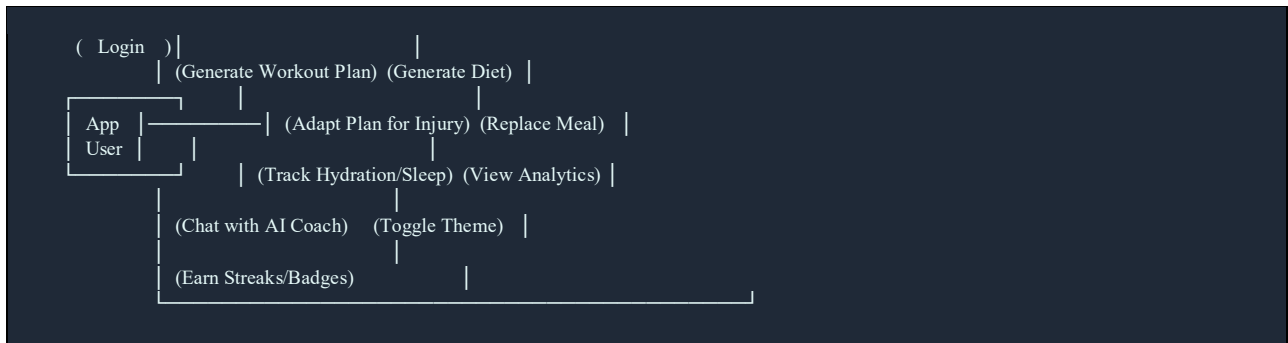


Figure 14.1 – Use Case Diagram for FitGenie AI

14.1 Use Case Descriptions

Use Case	Description
Register / Login	User creates an account or signs in
Complete Onboarding	User provides bio-metrics, goals, lifestyle, and medical information
Generate Workout/Diet Plan	System computes a personalized workout and nutrition plan
Adapt Plan for Injury	User flags a sore joint/muscle; system reroutes the workout plan
Replace Meal	User requests an alternative meal; groceries update automatically
Track Metrics / View Analytics	User logs hydration/sleep and views progress charts
Chat with AI Coach	User interacts with the conversational coaching panel

15. Data Flow Diagram (Level 0)

The Level-0 (context-level) DFD below presents FitGenie AI as a single process, showing the User and the major data flows into and out of the system.

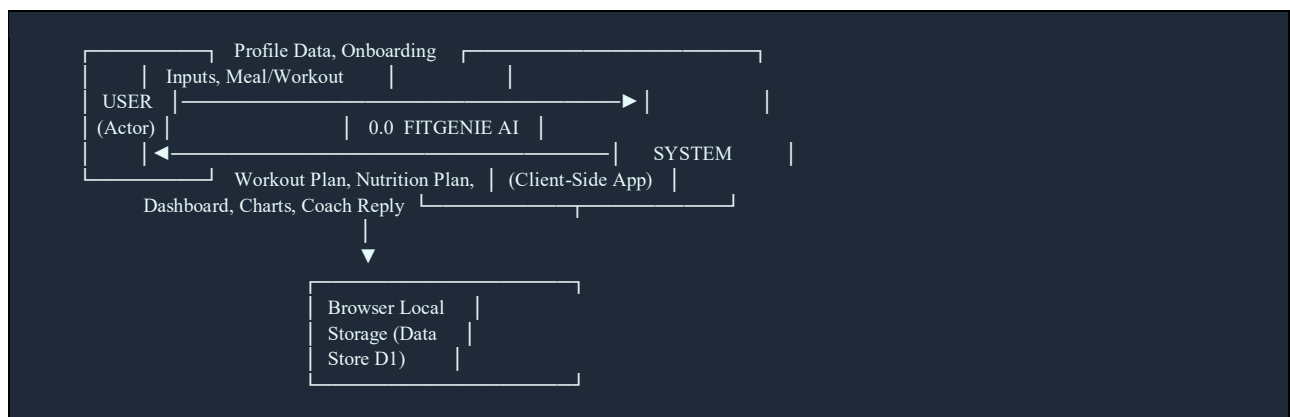


Figure 15.1 – Level 0 Data Flow Diagram

16. Working Methodology

The project followed an iterative, feature-driven methodology inspired by Agile principles, adapted to a solo-developer internship setting.

9. Requirement Gathering: Studied existing fitness apps to identify pain points and defined the feature list.
10. Wireframing & UI Planning: Sketched the landing page, onboarding wizard, and dashboard layout around a glassmorphic visual identity.
11. Project Setup: Initialised Git, created base HTML/CSS/JS files, and configured VS Code.
12. Landing Page Development: Built the marketing page (hero, features, testimonials, pricing, FAQ) first to establish the visual language.
13. Authentication Screens: Built Sign In/Up/Forgot Password with floating labels and inline validation.
14. Onboarding Wizard: Built the 4-step flow with choice-card selectors, sliders, and the live Bio-Calculator.
15. Recommendation Engine: Implemented BMR/TDEE (Mifflin-St Jeor) and workout/nutrition plan generation logic.
16. Dashboard & Feature Modules: Built the Dashboard, Workout, Nutrition, Grocery, Water, Metrics, and Progress modules.
17. Injury Adaptation Logic: Implemented the interactive body-model selector and exercise-swap rules.
18. Analytics & Gamification: Integrated Chart.js, canvas-confetti, streaks, and badges.
19. Responsive Styling & Testing: Reworked layouts for all breakpoints and ran functional/UI/browser tests, fixing bugs found.
20. Version Control & Deployment: Committed to GitHub and connected the repo to Netlify for continuous deployment.

17. Implementation

This chapter summarises how each major module of FitGenie AI was implemented, referencing the actual structure of the deployed application.

17.1 Landing Page

A fixed glass-effect navbar with anchor links leads into a hero section (animated progress ring and floating glass widgets), a six-card Features grid, a "How It Works" timeline, a testimonials slider, a three-tier pricing section, an FAQ accordion, and a closing call-to-action.

17.2 Login

Three interchangeable cards — Login, Signup, Forgot Password — toggle without a page reload, using floating-label inputs, real-time validation, and a password-visibility toggle.

17.3 Onboarding Wizard

A split layout pairs a live "Bio-Calculator" panel (BMR, TDEE, target calories, recalculated on every keystroke) with a 4-step form: Step 1 (age, gender, height, weight), Step 2 (goal, level, equipment, training days), Step 3 (diet, allergies, sleep/water sliders, activity level), and Step 4 (review, injuries, medical conditions).

17.4 Dashboard

The authenticated home screen surfaces daily targets (calories, water, workout status), the current streak, and shortcuts into Workout, Nutrition, and Progress — reusing the landing page's glass-card visual language.

17.5 Workout Plan

Renders the exercise structure generated from the user's profile, organised by training day with sets/reps/rest appropriate to the selected goal, and wired to the Injury Adaptation logic.

17.6 Nutrition

Displays computed macro targets and suggested meals respecting diet style and allergies, each with a "Replace Meal" action that regenerates an alternative meeting the same targets.

17.7 Grocery List

An interactive checklist auto-compiled from the current nutrition plan's ingredients, updating whenever a meal is replaced or regenerated.

17.8 Water Tracking & Daily Metrics

Users log fluid intake against their onboarding target via a visual fill indicator, while the Daily Metrics panel aggregates steps, active calories, sleep, and workout completion.

17.9 Progress Cards

Summarise weekly/streak-level achievements and feed the Chart.js analytics views for both short-term and 12-week-projection tracking.

18. Testing

18.1 Functional Testing

Test Case	Expected Result	Status
Signup validation	Inline error shown on invalid email; submission blocked	Pass
Login flow	User redirected to onboarding/dashboard	Pass
Live BMR calculation	BMR/TDEE/target calories update instantly	Pass
Workout plan generation	Plan reflects chosen goal/equipment/days	Pass
Meal replacement	New meal appears; grocery list updates	Pass
Injury adaptation	Affected exercises swapped for alternatives	Pass
Theme toggle	UI switches between light and dark mode	Pass

18.2 UI & Browser Compatibility Testing

- Verified consistent glassmorphic styling, spacing, and floating-label behaviour across all screens.
- Checked hover/lift micro-interactions, the testimonials slider, and the FAQ accordion.
- Tested on Chrome, Firefox, Edge, and Safari — fully functional on all, with minor backdrop-blur fallback styling added for Firefox/Safari.

18.3 Deployment Testing

- Verified every push to the GitHub main branch triggers a successful Netlify build and deploy.
- Confirmed the production URL loads all assets without 404/CORS errors.
- Tested the live URL on desktop and mobile to confirm parity with the local build.

19. Challenges Faced

19.1 Landing Page Overlapping Issue

Floating widget cards overlapped the hero illustration at tablet widths due to fixed pixel offsets. Resolved by switching to percentage-based positioning with CSS clamp() and dedicated tablet breakpoints.

19.2 Responsive Layout Issues

Multi-column sections (features grid, pricing, onboarding choice-cards) collapsed awkwardly on mobile. Resolved with auto-fit/minmax() grid templates and iterative testing across breakpoints in DevTools.

19.3 GitHub Repository Setup

Editor-specific files were accidentally committed. Resolved by adding a proper .gitignore and adopting a consistent commit-message convention.

19.4 Netlify Deployment

Early deploys failed silently due to a mismatched publish directory. Resolved by pointing Netlify's publish directory to the repository root and verifying via the deploy log before promoting to production.

19.5 Navigation & Dashboard Rendering

Screen-switching occasionally left stale event listeners, causing duplicate actions, and the dashboard sometimes rendered before plan data was ready. Both were resolved by fully re-rendering templates before re-binding listeners and adding guard checks/default states in rendering functions.

20. Results

The final outcome is a fully functional, publicly deployed web application meeting all objectives defined in Chapter 3, live at <https://fitgenie-ai.netlify.app> with source at <https://github.com/dhruv-jangid/fitgenie-ai>.

Objective (Chapter 3)	Achieved?
Fully responsive web application	Yes
Multi-step onboarding wizard	Yes
Live BMR/TDEE calculator	Yes
Personalized workout & nutrition plan generation	Yes
Interactive injury adaptation	Yes
Analytics dashboard & gamification	Yes
AI chat coach interface	Yes
Git/GitHub/Netlify professional workflow	Yes
Thorough functional/UI/cross-browser testing	Yes

21. Future Enhancements

- AI Chatbot: integrate a real LLM API for dynamic, context-aware coaching grounded in the user's history.
- Smartwatch Integration: pull step count, heart rate, and sleep data from wearable APIs (HealthKit, Google Fit, Fitbit).
- Voice Assistant: hands-free logging and coach queries via the Web Speech API.
- Meal Image Recognition: log meals by photo using an image-recognition model.
- Exercise Detection Using Camera: pose-estimation-based rep counting and form feedback.
- AI Injury Prediction: flag elevated injury risk from training load and soreness history before it occurs.
- A dedicated backend and database for real multi-device account sync and real payment processing.

22. Conclusion

FitGenie AI demonstrates that a thoughtfully designed, purely client-side web application — built with only HTML, CSS, and JavaScript — can deliver a genuinely personalized, engaging fitness and nutrition coaching experience. By focusing complexity on a rich onboarding data model and a computational recommendation engine rather than a heavy backend, the project stayed within an internship timeframe while remaining deployable and real-world usable.

Its value lies not only in the feature set but in the complete development lifecycle exercised — requirement analysis, UI/UX design, iterative implementation, systematic testing, and a genuine Git/GitHub/Netlify

deployment pipeline — mirroring practices used by professional front-end teams. The final deliverable, live at <https://fitgenie-ai.netlify.app> with an open codebase at <https://github.com/dhruv-jangid/fitgenie-ai>, stands as tangible evidence of the skills developed during this internship.

23. Learning Outcomes

- Framework-free JavaScript application structuring, including template-based view management and state handling.
- Modern CSS techniques — Grid, Flexbox, custom properties, glassmorphism, and keyframe animation.
- Translating domain knowledge (BMR/TDEE formulas) into working application logic.
- Integrating third-party CDN libraries (Chart.js, Lucide, canvas-confetti) into a vanilla JS project.
- Cross-browser/device testing methodology using DevTools.
- Real-world Git practices and end-to-end GitHub-to-Netlify continuous deployment.
- Requirement scoping, debugging discipline, time management, and technical documentation skills.

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